



MS/MA IN ECONOMICS

DYNAMIC STOCHASTIC MODELS

Final Exam

Hansen economy with CES utility function

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The Question

This exam solves for the impulse response functions of a *Hansen economy with a CES utility function* in which the elasticity of substitution in consumption is inelastic and the part of utility on leisure is logarithmic (indivisible labor). The driving force is a positive shock to the growth rate of the money supply. The question is:

Find the impulse response functions (graph them) for a Hansen economy with a CES utility function where the elasticity of substitution on consumption is inelastic. The part of the utility function on leisure is log. The shock is a positive money supply growth rate shock. Show all your work, including printouts of the computer programs you used and the impulse response function. Explain what you did.

Following the notation and development of McCandless (2008), *The ABCs of RBCs*, the tasks are:

1. Write out the model (households, firms, government) and its first-order conditions.
2. Compute the stationary state.
3. Write out the log-linear version of the model.
4. Write out the matrices $A, \dots, N$ that put the model in Uhlig's canonical form.
5. Use `llinsolve.m` to obtain the P, Q, R, S matrices and `impres.m` to graph the impulse response functions to a money-growth (seigniorage) shock.

The model is the cash-in-advance economy with seigniorage of Chapter 8 (Section 8.5), specialized to the CES utility function of the Appendix in Section 8.7. The real side is Hansen's indivisible-labor RBC of Chapter 6 (Section 6.4). The solution method (log-linearization, canonical matrices, and the matrix quadratic solved by generalized eigenvalues) follows Chapter 2 and Uhlig's method, implemented in the course routines `llinsolve.m` and `impres.m`. The logarithmic-utility version of exactly this model is the course program `mpsolve.m`, which is the $\eta = 1$ special case of what follows and serves as my benchmark. I set $\eta = 1.5$ (inelastic).

Notation. A hat denotes normalization by the money stock, $\hat{x}_t \equiv x_t/M_t$; in particular $\varphi_t \equiv M_t/M_{t-1}$ is the gross money growth rate, $\hat{p}_t \equiv p_t/M_t$ the normalized price level, and $\hat{g}_t$ the shock to the government deficit. A bar ($\bar{x}_t$) denotes a stationary-state value and a tilde ($\tilde{x}_t$) a log-deviation from it.

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1 The Model

1.1 The Monetary Environment

Money enters through a cash-in-advance (CIA) friction, following the Cooley–Hansen formulation of McCandless (2008, Sec. 8.1). The consumption good cannot be bought on credit: purchases in period t must be paid for with money the household carried out of period $t - 1$. With p_t the nominal price level and m_{t-1}^i the nominal money the household brings into the period, the cash-in-advance constraint is

$$p_t c_t^i \leq m_{t-1}^i \quad (1)$$

Money is thus an asset the household chooses to hold: it yields no interest but is needed to finance consumption, so real balances m_t^i/p_t enter the uses side of the flow budget constraint alongside physical capital,

$$k_{t+1}^i + \frac{m_t^i}{p_t} = w_t h_t^i + r_t k_t^i + (1 - \delta)k_t^i \quad (2)$$

This is the term m_t/p_t that appears on the left-hand side of the budget constraint at the start of Chapter 8: out of current income the household chooses how much to carry forward as capital and how much as real money balances.

Normalization. Because the money stock grows at the gross rate $\varphi_t \equiv M_t/M_{t-1} > 1$, nominal variables inherit this trend and have no stationary value. I normalize them by dividing by the current money stock, defining

$$\hat{m}_t^i \equiv \frac{m_t^i}{M_t} \quad \hat{p}_t \equiv \frac{p_t}{M_t} \quad (3)$$

Dividing the cash-in-advance constraint (1) by M_t and using

$$m_{t-1}^i/M_t = (m_{t-1}^i/M_{t-1})(M_{t-1}/M_t) = \hat{m}_{t-1}^i/\varphi_t$$

gives the normalized constraint

$$\hat{p}_t c_t^i \leq \frac{\hat{m}_{t-1}^i}{\varphi_t} \quad (4)$$

and dividing the budget constraint (2) by M_t , so that $m_t^i/p_t = \hat{m}_t^i/\hat{p}_t$, gives

$$k_{t+1}^i + \frac{\hat{m}_t^i}{\hat{p}_t} = w_t h_t^i + r_t k_t^i + (1 - \delta)k_t^i \quad (5)$$

The cash-in-advance constraint holds with equality whenever $\varphi_t > \beta$, since then carrying unspent money into the next period is dominated by consuming it now; this condition holds throughout, so (4) binds.

1.2 The Household

The representative household chooses $\{c_t, h_t, k_{t+1}, \hat{m}_t\}_{t=0}^{\infty}$ to maximize expected discounted utility with a CES felicity in consumption and an indivisible-labor (linear) term in hours,

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(c_t, 1 - h_t) \quad u(c_t, 1 - h_t) = \frac{c_t^{1-\eta} - 1}{1-\eta} + B h_t \quad (6)$$

where $\eta > 0$ governs the intertemporal elasticity of substitution in consumption, $1/\eta$. The exam requires an inelastic elasticity, $1/\eta < 1 \Leftrightarrow \eta > 1$; I set $\eta = 1.5$ (Section 4.1 shows why η must stay below a determinacy threshold). As $\eta \rightarrow 1$ the first term tends to $\ln c_t$ and the model collapses to the logarithmic economy solved by `mpsolve.m`. The linear term $B h_t$ is the aggregate counterpart of Hansen's indivisible labor: an individual has logarithmic utility of leisure $\ln(1 - h)$, and after the employment-lottery aggregation the representative household's felicity becomes linear in hours with

$$B = \frac{A \ln(1 - h_0)}{h_0} < 0 \quad (7)$$

Why indivisible labor. Households cannot vary hours on the intensive margin but only decide whether to work a fixed shift, and an employment lottery convexifies the choice. Aggregating over the lottery turns the individual's logarithmic leisure $\ln(1 - h)$ into a felicity that is linear in aggregate hours, with the constant B of (7). Two reasons make this the right specification here. First, it is what "Hansen economy" and "the part on leisure is log" mean, and it is the form on which the book builds the cash-in-advance and CES apparatus (McCandless, 2008, Sec. 6.4, 8.5, 8.7). Second, the linear term makes the marginal disutility of work constant, so aggregate labor supply is highly elastic; this is the channel that lets a purely monetary disturbance move the real economy at all, and it is the empirically motivated feature (most cyclical labor-market adjustment occurs on the extensive margin) that Hansen introduced to amplify the business cycle.

Following the recursive route of McCandless (2008, Sec. 8.5.1): the two binding constraints (4)–(5) are solved for c_t and h_t ,

$$c_t = \frac{\hat{m}_{t-1}}{\varphi_t \hat{p}_t} \quad h_t = \frac{k_{t+1} + \hat{m}_t / \hat{p}_t - r_t k_t - (1 - \delta) k_t}{w_t} \quad (8)$$

and substituted into (6), turning the problem into an unconstrained maximization over the two genuine choices $\{\hat{m}_t, k_{t+1}\}$,

$$\max_{\{\hat{m}_t, k_{t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{\left(\frac{\hat{m}_{t-1}}{\varphi_t \hat{p}_t} \right)^{1-\eta} - 1}{1-\eta} + B \frac{k_{t+1} + \hat{m}_t / \hat{p}_t - r_t k_t - (1 - \delta) k_t}{w_t} \right] \quad (9)$$

First-order condition for capital. The choice k_{t+1} appears in (9) twice: inside h_t in the period- t term (through $+k_{t+1}/w_t$) and inside h_{t+1} in the period- $(t+1)$ term (through $-(r_{t+1} + 1 - \delta)k_{t+1}/w_{t+1}$). Consumption does not depend on k_{t+1} , so only the linear labor term contributes. Differentiating and setting the sum to zero,

$$\beta^t B \frac{1}{w_t} + \beta^{t+1} \mathbb{E}_t \left[B \frac{-(r_{t+1} + 1 - \delta)}{w_{t+1}} \right] = 0 \quad (10)$$

Dividing by $\beta^t B$ and rearranging gives the capital first-order condition

$$\frac{1}{w_t} = \beta \mathbb{E}_t \left[\frac{r_{t+1} + 1 - \delta}{w_{t+1}} \right] \quad (11)$$

which carries no η because it comes entirely from the linear term Bh .

First-order condition for money. The choice $\hat{m}_t$ also appears twice: inside h_t in the period- t term (through $+\hat{m}_t/(\hat{p}_t w_t)$) and inside c_{t+1} in the period- $(t+1)$ term, since $c_{t+1} = \hat{m}_t/(\varphi_{t+1} \hat{p}_{t+1})$. The marginal utility of consumption is $\partial u/\partial c = c^{-\eta}$ and $\partial c_{t+1}/\partial \hat{m}_t = 1/(\varphi_{t+1} \hat{p}_{t+1})$, so

$$\beta^t B \frac{1}{\hat{p}_t w_t} + \beta^{t+1} \mathbb{E}_t \left[c_{t+1}^{-\eta} \frac{1}{\varphi_{t+1} \hat{p}_{t+1}} \right] = 0 \quad (12)$$

In equilibrium normalized money aggregates to $\hat{m}_t = 1$ (Section 1.5), so $c_{t+1} = 1/(\varphi_{t+1} \hat{p}_{t+1})$ and $c_{t+1}^{-\eta} = (\varphi_{t+1} \hat{p}_{t+1})^\eta$. Substituting, $c_{t+1}^{-\eta}/(\varphi_{t+1} \hat{p}_{t+1}) = (\varphi_{t+1} \hat{p}_{t+1})^{\eta-1}$, and dividing (12) by β^t gives the money first-order condition

$$\boxed{\beta \mathbb{E}_t \left[(\varphi_{t+1} \hat{p}_{t+1})^{\eta-1} \right] = -\frac{B}{\hat{p}_t w_t}} \quad (13)$$

Why this differs from the logarithmic case. With logarithmic utility the marginal utility of consumption is $1/c$; McCandless (2008, Sec. 8.2, p. 191) obtains the money condition $B/(w_t \hat{p}_t) = -\beta \mathbb{E}_t[1/(\hat{p}_{t+1} c_{t+1} \varphi_{t+1})]$. Because $c_{t+1} = 1/(\varphi_{t+1} \hat{p}_{t+1})$ in equilibrium, the factor $1/c_{t+1} = \varphi_{t+1} \hat{p}_{t+1}$ exactly cancels, collapsing that condition to the *static* relation $B/(w_t \hat{p}_t) = -\beta$. Under CES the marginal utility is instead $c_{t+1}^{-\eta} = (\varphi_{t+1} \hat{p}_{t+1})^\eta$, which does not fully cancel and leaves the term $(\varphi_{t+1} \hat{p}_{t+1})^{\eta-1}$ inside the expectation. The single difference between (13) and the logarithmic money condition is therefore the replacement of $1/c$ by $c^{-\eta}$: it is exactly the CES curvature in consumption that turns the money condition into a genuinely expectational equation.

1.3 The Firm

Perfectly competitive firms operate a Cobb–Douglas technology,

$$Y_t = \lambda_t K_t^\theta H_t^{1-\theta} \quad (14)$$

with $\theta \in (0,1)$ the capital share and λ_t total factor productivity. Profit maximization equates each factor price to its marginal product,

$$r_t = \theta \lambda_t K_t^{\theta-1} H_t^{1-\theta} \quad (15)$$

$$w_t = (1 - \theta) \lambda_t K_t^\theta H_t^{-\theta} \quad (16)$$

and zero profits imply $Y_t = w_t H_t + r_t K_t$. Productivity follows the autoregressive process

$$\ln \lambda_t = \gamma \ln \lambda_{t-1} + \varepsilon_t^\lambda, \quad 0 < \gamma < 1, \quad \bar{\lambda} = 1 \quad (17)$$

1.4 The Government and Seigniorage

The government purchases $g_t = \hat{g}_t \bar{g}$ units of real goods each period and finances them entirely by printing money (McCandless, 2008, Sec. 8.5.1, p. 212):

$$g_t = \hat{g}_t \bar{g} = \frac{M_t - M_{t-1}}{p_t} \quad (18)$$

Dividing numerator and denominator by M_t and using $\varphi_t = M_t/M_{t-1}$ gives the normalized government budget constraint,

$$\hat{p}_t \hat{g}_t \bar{g} = 1 - \frac{1}{\varphi_t} \quad (19)$$

where the deficit shock follows

$$\ln \hat{g}_t = \pi \ln \hat{g}_{t-1} + \varepsilon_t^g, \quad \bar{g} = 1 \quad (20)$$

Equation (19) is the seigniorage (inflation-tax) relationship: a larger deficit ($\varepsilon_t^g > 0$) must be financed by faster money creation, so the money growth rate φ_t rises. The exogenous deficit shock therefore is the positive money-supply growth-rate shock, transmitted one-for-one to φ_t .

1.5 First-Order and Equilibrium Conditions

All households in the unit mass are identical, so individual and aggregate variables coincide, $c_t = C_t$, $k_{t+1} = K_{t+1}$, $h_t = H_t$, and normalized money aggregates to $\hat{m}_t = M_t/M_t = 1$. The economy-wide cash-in-advance constraint states that consumption is paid with money carried from the previous period, which after normalization becomes $\hat{p}_t C_t = 1/\varphi_t$. Collecting the household optimality conditions (11)–(13), the firm conditions (15)–(16), the resource constraint, and the government bud-

get (19), the equilibrium is described by

$$\hat{p}_t C_t = \frac{1}{\varphi_t} \quad (21)$$

$$\hat{p}_t \hat{g}_t \bar{g} = 1 - \frac{1}{\varphi_t} \quad (22)$$

$$C_t + K_{t+1} + \hat{g}_t \bar{g} = w_t H_t + r_t K_t + (1 - \delta) K_t \quad (23)$$

$$r_t = \theta \lambda_t K_t^{\theta-1} H_t^{1-\theta} \quad (24)$$

$$w_t = (1 - \theta) \lambda_t K_t^\theta H_t^{-\theta} \quad (25)$$

$$\frac{1}{w_t} = \beta \mathbb{E}_t \left[\frac{r_{t+1} + 1 - \delta}{w_{t+1}} \right] \quad (26)$$

$$\beta \mathbb{E}_t \left[(\varphi_{t+1} \hat{p}_{t+1})^{\eta-1} \right] = - \frac{B}{\hat{p}_t w_t} \quad (27)$$

together with the exogenous processes (17) and (20). Adding (21) and (22) gives $C_t + \hat{g}_t \bar{g} = 1/\hat{p}_t$, so the resource constraint (23) can be written compactly as

$$K_{t+1} + \frac{1}{\hat{p}_t} = w_t H_t + r_t K_t + (1 - \delta) K_t \quad (28)$$

Equation (21) only determines C_t as a function of $\hat{p}_t$ and φ_t ; following McCandless (2008, p. 226) I set it aside when log-linearizing and recover consumption from it afterward. The remaining six equations (22)–(27) in the six endogenous variables $\{K_{t+1}, \varphi_t, r_t, w_t, \hat{p}_t, H_t\}$ are the system I take to the stationary state and then log-linearize.

1.6 The Stationary State

In a stationary state every variable is constant, $X_t = \bar{X}$ for all t . The exogenous processes (17) and (20) then imply $\bar{\lambda} = 1$ and $\bar{g} = 1$. Imposing constancy on the equilibrium conditions (21)–(27) yields the stationary system

$$\bar{p} \bar{C} = \frac{1}{\bar{\varphi}} \quad (29)$$

$$\bar{p} \bar{g} = 1 - \frac{1}{\bar{\varphi}} \quad (30)$$

$$\bar{K} + \frac{1}{\bar{p}} = \bar{w} \bar{H} + \bar{r} \bar{K} + (1 - \delta) \bar{K} \quad (31)$$

$$\bar{r} = \theta \bar{K}^{\theta-1} \bar{H}^{1-\theta} \quad (32)$$

$$\bar{w} = (1 - \theta) \bar{K}^\theta \bar{H}^{-\theta} \quad (33)$$

$$1 = \beta (\bar{r} + 1 - \delta) \quad (34)$$

$$\beta (\bar{\varphi} \bar{p})^{\eta-1} = - \frac{B}{\bar{p} \bar{w}} \quad (35)$$

The system is solved recursively. The capital first-order condition (34) gives the net return on capital,

$$\bar{r} = \frac{1}{\beta} - 1 + \delta \quad (36)$$

which is independent of η . The factor-price conditions (32)–(33) then deliver the capital–labor ratio and the wage,

$$\frac{\bar{K}}{\bar{H}} = \left(\frac{\theta}{\bar{r}} \right)^{\frac{1}{1-\theta}}, \quad \bar{w} = (1 - \theta) \left(\frac{\theta}{\bar{r}} \right)^{\frac{\theta}{1-\theta}} \quad (37)$$

The money first-order condition (35), the only relation involving η , is solved for the normalized price. Multiplying by $\bar{p} \bar{w}$ and collecting $\bar{p}$,

$$\beta \bar{w} \bar{\varphi}^{\eta-1} \bar{p}^\eta = -B \implies \bar{p} = \left(\frac{-B}{\bar{w} \beta} \right)^{1/\eta} \bar{\varphi}^{\frac{1}{\eta}-1} \quad (38)$$

Because $B < 0$, the base $-B$ is positive and $\bar{p}$ is well defined. The cash-in-advance constraint (29) gives consumption,

$$\bar{C} = \frac{1}{\bar{p} \bar{\varphi}} = \left(\frac{-\bar{w} \beta}{B} \right)^{1/\eta} \bar{\varphi}^{-1/\eta} \quad (39)$$

and the government budget (30) gives real public spending,

$$\bar{g} = \frac{1 - 1/\bar{\varphi}}{\bar{p}} \quad (40)$$

Finally, the compact resource constraint (31) reads $1/\bar{p} = \bar{w} \bar{H} + (\bar{r} - \delta) \bar{K}$; writing $\bar{K} = (\bar{K}/\bar{H}) \bar{H}$ and factoring out $\bar{H}$ yields hours, capital, and output,

$$\bar{H} = \frac{1/\bar{p}}{\bar{w} + (\bar{r} - \delta) \bar{K}/\bar{H}}, \quad \bar{K} = \frac{\bar{K}}{\bar{H}} \bar{H}, \quad \bar{Y} = \bar{K}^\theta \bar{H}^{1-\theta} \quad (41)$$

together with $\bar{m} = 1$. Equations (38)–(39) show the role of the inflation tax: since $\bar{C} \propto \bar{\varphi}^{-1/\eta}$ while $\bar{H}, \bar{Y} \propto \bar{\varphi}^{1-1/\eta}$, faster money growth taxes cash-financed consumption and shifts activity toward hours and output (McCandless, 2008, Sec. 8.7, p. 226 and Fig. 8.6). Numerical values for the calibration $\eta = 1.5$ are reported in Table 2 of Section 4.

2 The Log-Linear Version

2.1 Uhlig's Method

I log-linearize the model around the stationary state using Uhlig's method. For any positive variable X_t I write

$$X_t = \bar{X} e^{\tilde{X}_t}, \quad \tilde{X}_t \approx \frac{X_t - \bar{X}}{\bar{X}} \quad (42)$$

so that $\tilde{X}_t$ is the log-deviation of X_t from its stationary value (in particular, $\tilde{p}_t$ denotes the log-deviation of the normalized price $\hat{p}_t$). Using the first-order approximation $e^{\tilde{X}_t} \approx 1 + \tilde{X}_t$ and dropping products of deviations, two rules follow:

- a product of powers linearizes into a weighted sum of deviations, with the exponents as weights, $A_t^\alpha B_t^\beta \rightarrow \alpha \tilde{A}_t + \beta \tilde{B}_t$;
- a sum linearizes into deviations weighted by stationary levels, $A_t + B_t \rightarrow \bar{A} \tilde{A}_t + \bar{B} \tilde{B}_t$.

In each case the stationary-state relation cancels the constant terms.

2.2 Intratemporal Conditions

The rental rate (24) and the wage (25) are products of powers, so applying the first rule directly gives

$$\tilde{r}_t = \tilde{\lambda}_t + (\theta - 1)\tilde{K}_t + (1 - \theta)\tilde{H}_t \quad (43)$$

$$\tilde{w}_t = \tilde{\lambda}_t + \theta\tilde{K}_t - \theta\tilde{H}_t \quad (44)$$

For the government budget (22), substituting $\hat{p}_t \hat{g}_t \bar{g} = \tilde{p} \bar{g} e^{\tilde{p}_t + \tilde{g}_t}$ (with $\bar{g} = 1$) on the left and $1 - \varphi_t^{-1} = 1 - \bar{\varphi}^{-1} e^{-\tilde{\varphi}_t}$ on the right, and cancelling constants through the stationary relation $\tilde{p} \bar{g} = 1 - \bar{\varphi}^{-1}$,

$$0 = -\frac{1}{\bar{\varphi}} \tilde{\varphi}_t + \tilde{p} \bar{g} \tilde{p}_t + \tilde{p} \bar{g} \tilde{g}_t \quad (45)$$

2.3 The Resource Constraint

Starting from the compact resource constraint (28) and applying the sum rule (each level term weighted by its stationary value, with $1/\hat{p}_t$ contributing $-\tilde{p}^{-1} \tilde{p}_t$ because its exponent is -1), the constant terms cancel by the stationary constraint $\bar{K} + \tilde{p}^{-1} = \bar{w} \bar{H} + \bar{r} \bar{K} + (1 - \delta) \bar{K}$. Collecting the coefficient of $\tilde{K}_t$, namely $\bar{r} \bar{K} + (1 - \delta) \bar{K} = (\bar{r} + 1 - \delta) \bar{K}$,

$$0 = \bar{K} \tilde{K}_{t+1} - (\bar{r} + 1 - \delta) \bar{K} \tilde{K}_t - \bar{r} \bar{K} \tilde{r}_t - \bar{w} \bar{H} \tilde{w}_t - \frac{1}{\bar{p}} \tilde{p}_t - \bar{w} \bar{H} \tilde{H}_t \quad (46)$$

2.4 The Euler Equations

Capital. Starting from (26), the left side gives $-\tilde{w}_t$, the numerator $r_{t+1} + 1 - \delta$ linearizes into $(\bar{r} + 1 - \delta) + \bar{r} \tilde{r}_{t+1}$, and $1/w_{t+1}$ into $-\tilde{w}_{t+1}$. The stationary relation $\bar{r} + 1 - \delta = 1/\bar{\beta}$ cancels the constants and makes $\beta(\bar{r} + 1 - \delta) = 1$; multiplying through by $\bar{w}$,

$$0 = \mathbb{E}_t[\beta \bar{r} \tilde{r}_{t+1} - \tilde{w}_{t+1} + \tilde{w}_t] \quad (47)$$

This condition carries no η : it is identical to the logarithmic case.

Money. Starting from the CES money condition (27), the left side linearizes as $(\bar{\varphi}\bar{p})^{\eta-1}[1 + (\eta - 1)(\tilde{\varphi}_{t+1} + \tilde{p}_{t+1})]$ and the right side as $-B(\bar{p}\bar{w})^{-1}(1 - \tilde{p}_t - \tilde{w}_t)$. Cancelling constants through the stationary relation $\beta(\bar{\varphi}\bar{p})^{\eta-1} = -B/(\bar{p}\bar{w})$ and dividing by that common positive constant,

$$0 = \mathbb{E}_t[(\eta - 1)(\tilde{\varphi}_{t+1} + \tilde{p}_{t+1}) + \tilde{w}_t + \tilde{p}_t] \quad (48)$$

Equation (48) is the only place where η enters the log-linear model. At $\eta = 1$ it degenerates to the static relation $\tilde{p}_t = -\tilde{w}_t$ of the logarithmic economy; for $\eta > 1$ it is a genuine expectational equation carrying $\tilde{\varphi}_{t+1}$ and $\tilde{p}_{t+1}$, which is why it joins the second block (McCandless, 2008, Sec. 8.7, p. 228).

2.5 The Exogenous Processes

Since $\bar{\lambda} = \bar{g} = 1$, the log-deviations coincide with the log-levels, and the technology process (17) and the deficit process (20) are already linear,

$$\tilde{\lambda}_t = \gamma\tilde{\lambda}_{t-1} + \varepsilon_t^\lambda, \quad \tilde{g}_t = \pi\tilde{g}_{t-1} + \varepsilon_t^g \quad (49)$$

2.6 The Log-Linearized System

Collecting (43)–(49), the model reduces to four static equations,

$$0 = -\frac{1}{\bar{\varphi}}\tilde{\varphi}_t + \bar{p}\bar{g}\tilde{p}_t + \bar{p}\bar{g}\tilde{g}_t \quad (50)$$

$$0 = \bar{K}\tilde{K}_{t+1} - (\bar{r} + 1 - \delta)\bar{K}\tilde{K}_t - \bar{r}\bar{K}\tilde{r}_t - \bar{w}\bar{H}\tilde{w}_t - \frac{1}{\bar{p}}\tilde{p}_t - \bar{w}\bar{H}\tilde{H}_t \quad (51)$$

$$0 = \tilde{r}_t - \tilde{\lambda}_t + (1 - \theta)\tilde{K}_t - (1 - \theta)\tilde{H}_t \quad (52)$$

$$0 = \tilde{w}_t - \tilde{\lambda}_t - \theta\tilde{K}_t + \theta\tilde{H}_t \quad (53)$$

two expectational equations,

$$0 = \mathbb{E}_t[\beta\bar{r}\tilde{r}_{t+1} - \tilde{w}_{t+1} + \tilde{w}_t] \quad (54)$$

$$0 = \mathbb{E}_t[(\eta - 1)(\tilde{\varphi}_{t+1} + \tilde{p}_{t+1}) + \tilde{w}_t + \tilde{p}_t] \quad (55)$$

and the two exogenous processes (49). Consumption does not appear: it is recovered from the cash-in-advance relation (21), whose log-linear form is $\tilde{p}_t + \tilde{C}_t = -\tilde{\varphi}_t$. The static block (50)–(53) and the expectational block (54)–(55) are cast in matrix form in Section 3.

3 The Matrices for Solving the Model

3.1 State, Jump and Exogenous Vectors

Following Uhlig's method, the endogenous variables are sorted into jumps and states by a counting rule. The jump variables are determined within the period; there are as many of them as static equations (four), and they are recovered from the states through $y_t = Rx_{t-1} + Sz_t$. The state variables carry their own law of motion $x_t = Px_{t-1} + Qz_t$; there are as many of them as expectational equations (two), since each Euler equation governs the intertemporal dynamics of one state — the capital first-order condition that of capital, and the money first-order condition that of the money growth rate. Accordingly,

$$x_t = \begin{bmatrix} \tilde{K}_{t+1} \\ \tilde{\varphi}_t \end{bmatrix} \quad y_t = \begin{bmatrix} \tilde{r}_t \\ \tilde{w}_t \\ \tilde{p}_t \\ \tilde{H}_t \end{bmatrix} \quad z_t = \begin{bmatrix} \tilde{\lambda}_t \\ \tilde{g}_t \end{bmatrix} \quad (56)$$

with $x_{t-1} = [\tilde{K}_t, \tilde{\varphi}_{t-1}]'$ and $x_{t+1} = [\tilde{K}_{t+2}, \tilde{\varphi}_{t+1}]'$. The timing convention, in which capital enters the state vector as $\tilde{K}_{t+1}$, lets the capital Euler equation (which involves $\tilde{K}_{t+2}, \tilde{K}_{t+1}, \tilde{K}_t$) be written directly in terms of x_{t+1}, x_t, x_{t-1} .

3.2 The Canonical Form

The log-linearized model is written in Uhlig's canonical form,

$$0 = Ax_t + Bx_{t-1} + Cy_t + Dz_t \quad (57)$$

$$0 = \mathbb{E}_t[Fx_{t+1} + Gx_t + Hx_{t-1} + Jy_{t+1} + Ky_t + Lz_{t+1} + Mz_t] \quad (58)$$

$$z_{t+1} = Nz_t + \varepsilon_{t+1} \quad (59)$$

The first block (57) collects the four static equations — the government budget (50), the resource constraint (51), and the two factor-price conditions (52)–(53) — so it has four rows. The second block (58) collects the two expectational equations, the capital and money Euler equations (54)–(55), so it has two rows.

Solution method. The first block (57) has four static equations and four jump variables, and the coefficient matrix C on y_t is square and invertible (numerically $\det C = -0.2137 \neq 0$). This is the regular case of Uhlig's method: the jumps can be eliminated as $y_t = -C^{-1}(Ax_t + Bx_{t-1} + Dz_t)$, so the model reduces to a single matrix quadratic equation in the state transition P .

3.3 First-Block Matrices

Reading the coefficients of (50)–(53) against $x_t = [\tilde{K}_{t+1}, \tilde{\varphi}_t]'$, $x_{t-1} = [\tilde{K}_t, \tilde{\varphi}_{t-1}]'$, $y_t = [\tilde{r}_t, \tilde{w}_t, \tilde{p}_t, \tilde{H}_t]'$ and $z_t = [\tilde{\lambda}_t, \tilde{g}_t]'$, the matrices multiplying the state vectors are

$$A = \begin{bmatrix} 0 & -\frac{1}{\bar{\varphi}} \\ \bar{K} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ -(\bar{r} + 1 - \delta)\bar{K} & 0 \\ 1 - \theta & 0 \\ -\theta & 0 \end{bmatrix} \quad (60)$$

both of dimension 4×2 . The matrix multiplying the jump vector is

$$C = \begin{bmatrix} 0 & 0 & \bar{p} \bar{g} & 0 \\ -\bar{r}\bar{K} & -\bar{w}\bar{H} & -\frac{1}{\bar{p}} & -\bar{w}\bar{H} \\ 1 & 0 & 0 & -(1 - \theta) \\ 0 & 1 & 0 & \theta \end{bmatrix} \quad (4 \times 4) \quad (61)$$

and the matrix multiplying the exogenous vector is

$$D = \begin{bmatrix} 0 & \bar{p} \bar{g} \\ 0 & 0 \\ -1 & 0 \\ -1 & 0 \end{bmatrix} \quad (4 \times 2) \quad (62)$$

3.4 Second-Block Matrices

The second block collects the capital Euler equation (54) (first row) and the money Euler equation (55) (second row). Reading their coefficients against x_{t+1}, x_t, x_{t-1} gives

$$F = \begin{bmatrix} 0 & 0 \\ 0 & \eta - 1 \end{bmatrix} \quad G = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad H = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (63)$$

all 2×2 ; against the jump vectors y_{t+1}, y_t ,

$$J = \begin{bmatrix} \beta\bar{r} & -1 & 0 & 0 \\ 0 & 0 & \eta - 1 & 0 \end{bmatrix} \quad K = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} \quad (64)$$

both 2×4 ; and against the exogenous vectors z_{t+1}, z_t ,

$$L = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad M = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (65)$$

both 2×2 . The single appearance of η (in F and J , through $\eta - 1$) is the matrix footprint of the CES utility: at $\eta = 1$ these entries vanish and the money Euler equation degenerates to a static one.

3.5 Law of Motion of the Shocks

The exogenous block $z_{t+1} = Nz_t + \varepsilon_{t+1}$ stacks the technology process and the deficit process (49), with persistences γ and π ,

$$N = \begin{bmatrix} \gamma & 0 \\ 0 & \pi \end{bmatrix} \quad (2 \times 2) \quad \varepsilon_{t+1} = \begin{bmatrix} \varepsilon_{t+1}^\lambda \\ \varepsilon_{t+1}^g \end{bmatrix} \quad (66)$$

The money-supply growth-rate shock is the second innovation, ε^g (the second column of N); this is the shock fed through `impres.m` in Section 4.

4 Numerical Solution and Impulse Responses

4.1 Calibration and Stationary State

The model is calibrated following the program of McCandless (2008, Sec. 8.9), changing only the CES curvature to an inelastic value, $\eta = 1.5$ (intertemporal elasticity of substitution $1/\eta = 0.667 < 1$). Table 1 lists the parameters and Table 2 the resulting stationary state.

	θ	β	δ	γ	π	$\bar{\varphi}$	η	B
Value	0.36	0.99	0.025	0.95	0.48	1.3180	1.5	-2.5805

Table 1: Calibration.

	$\bar{r}$	$\bar{K}/\bar{H}$	$\bar{w}$	$\bar{p}$	$\bar{C}$	$\bar{g}$	$\bar{H}$	$\bar{K}$	$\bar{Y}$
Value	0.0351	37.9893	2.3706	0.9716	0.7809	0.2483	0.3737	14.1953	1.3841

Table 2: Stationary state for $\eta = 1.5$ (with $\bar{m} = \bar{\lambda} = \bar{g} = 1$).

Determinacy and the choice of η . The curvature η cannot be set arbitrarily large. The generalized-eigenvalue problem solved by `llinsolve.m` has a root equal to $1/\bar{\varphi}$; since $\bar{\varphi} > 1$ under positive money growth, this root falls inside the unit circle once η is large enough, yielding a third stable

root against only two state variables. The Blanchard–Kahn condition then fails and the equilibrium becomes indeterminate. The threshold is $\eta^* \approx 1.76$: for $\eta < \eta^*$ the model has a unique saddle-path solution, whereas for $\eta \geq \eta^*$ (in particular $\eta = 2$) it does not. The inelastic value $\eta = 1.5$ lies safely below the threshold. The book uses $\eta = 2$ only for the static steady-state utility comparison of its Figure 8.6, never for the dynamics.

4.2 Solution Matrices

Substituting the parameters of Table 1 and the stationary-state values of Table 2 into the matrices of Section 3 gives the numerical system,

$$A = \begin{bmatrix} 0 & -0.7587 \\ 14.1953 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ -14.3386 & 0 \\ 0.64 & 0 \\ -0.36 & 0 \end{bmatrix} \quad C = \begin{bmatrix} 0 & 0 & 0.2413 & 0 \\ -0.4983 & -0.8858 & -1.0292 & -0.8858 \\ 1 & 0 & 0 & -0.64 \\ 0 & 1 & 0 & 0.36 \end{bmatrix} \quad (67)$$

$$D = \begin{bmatrix} 0 & 0.2413 \\ 0 & 0 \\ -1 & 0 \\ -1 & 0 \end{bmatrix} \quad F = \begin{bmatrix} 0 & 0 \\ 0 & 0.5 \end{bmatrix} \quad J = \begin{bmatrix} 0.0348 & -1 & 0 & 0 \\ 0 & 0 & 0.5 & 0 \end{bmatrix} \quad N = \begin{bmatrix} 0.95 & 0 \\ 0 & 0.48 \end{bmatrix} \quad (68)$$

with G, H, L, M zero. Solving this system, `llinsolve.m` returns the recursive law of motion $x_t = Px_{t-1} + Qz_t$, $y_t = Rx_{t-1} + Sz_t$, with

$$P = \begin{bmatrix} 0.9418 & 0 \\ -0.1176 & 0 \end{bmatrix} \quad Q = \begin{bmatrix} 0.1355 & -0.0033 \\ -0.1163 & 0.3004 \end{bmatrix} \quad (69)$$

$$R = \begin{bmatrix} -1.0654 & 0 \\ 0.5993 & 0 \\ -0.3698 & 0 \\ -0.6646 & 0 \end{bmatrix} \quad S = \begin{bmatrix} 1.6621 & 0.0076 \\ 0.6276 & -0.0043 \\ -0.3656 & -0.0554 \\ 1.0345 & 0.0118 \end{bmatrix} \quad (70)$$

or, written out,

$$\tilde{K}_{t+1} = 0.9418 \tilde{K}_t + 0.1355 \tilde{\lambda}_t - 0.0033 \tilde{g}_t \quad (71)$$

$$\tilde{\varphi}_t = -0.1176 \tilde{K}_t - 0.1163 \tilde{\lambda}_t + 0.3004 \tilde{g}_t \quad (72)$$

$$\tilde{r}_t = -1.0654 \tilde{K}_t + 1.6621 \tilde{\lambda}_t + 0.0076 \tilde{g}_t \quad (73)$$

$$\tilde{w}_t = 0.5993 \tilde{K}_t + 0.6276 \tilde{\lambda}_t - 0.0043 \tilde{g}_t \quad (74)$$

$$\tilde{p}_t = -0.3698 \tilde{K}_t - 0.3656 \tilde{\lambda}_t - 0.0554 \tilde{g}_t \quad (75)$$

$$\tilde{H}_t = -0.6646 \tilde{K}_t + 1.0345 \tilde{\lambda}_t + 0.0118 \tilde{g}_t \quad (76)$$

The eigenvalues of P are 0.9418 and 0: capital is highly persistent, while the money-growth state has no intrinsic persistence and is driven entirely by the shocks (through Q). Consumption is recovered as $\tilde{C}_t = -\tilde{\varphi}_t - \tilde{p}_t$. As a check on the program, setting $\eta = 1$ reproduces the logarithmic economy of `mpsolve.m`, the special case of Section 1.2.

4.3 Impulse Responses

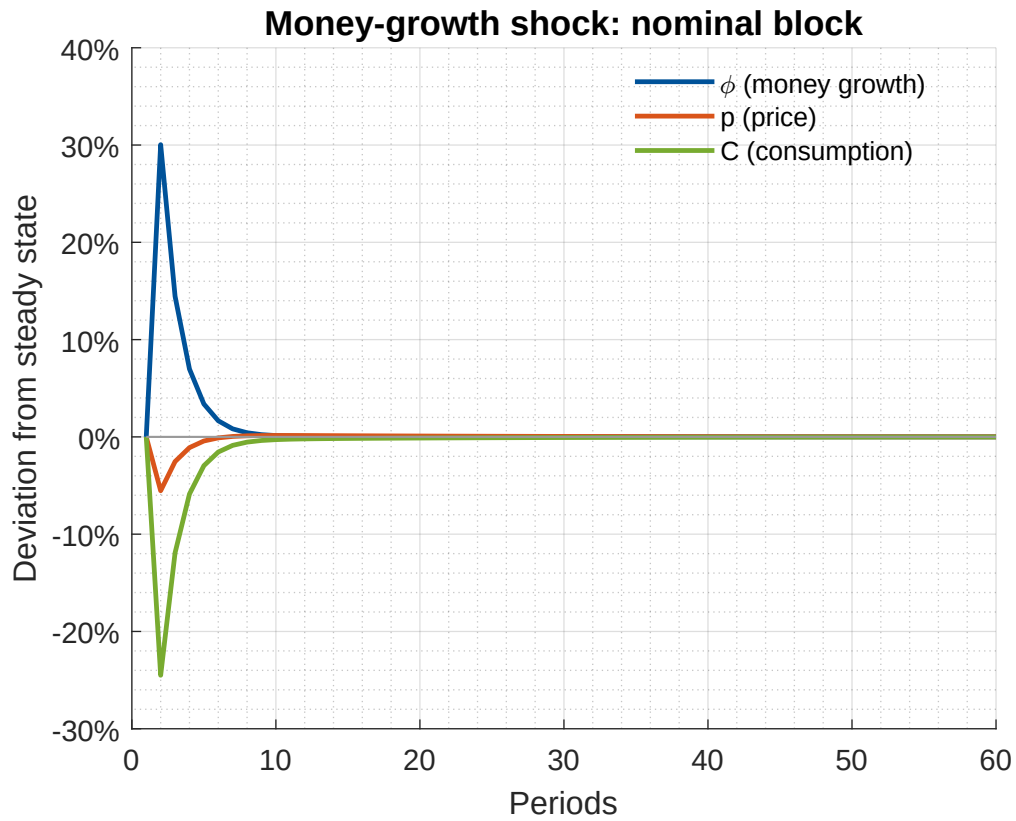
Figure 1 shows the impulse responses to a one-unit innovation in the deficit (money-growth) shock ε^8 — the positive money-supply growth-rate shock of the prompt. Nominal and real variables respond on very different scales, so they are shown in two panels.

On impact the shock raises the money growth rate φ by about 30%. Faster money creation is an inflation tax on the cash-financed consumption good: the normalized price $\hat{p} = p/M$ falls (the money stock grows faster than the price level) and consumption drops sharply, by roughly 24%, since $\tilde{C} = -\tilde{\varphi} - \tilde{p}$. The household reallocates toward work — hours rise by about 1.2% — which raises the marginal product of capital (the rental rate increases by 0.8%) and lowers the marginal product of labor (the wage falls by 0.4%). Capital declines gradually, reaching about -0.5% after five quarters as investment is crowded out. As the shock dies away (persistence $\pi = 0.48$) every variable returns monotonically to the stationary state, the slow tail being governed by the persistence of capital (0.94).

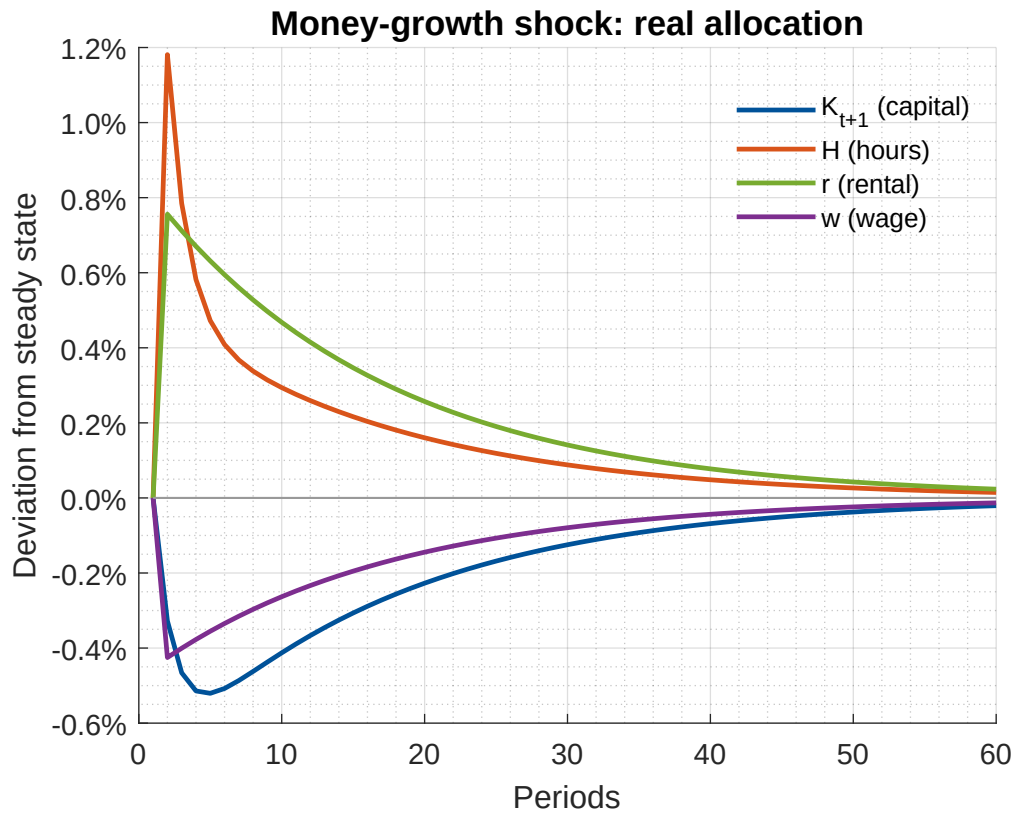
Reading the responses through the model. Three features of the model shape the picture above. First, the sharp impact drop of consumption is not a preference phenomenon but a *quantity* one: the cash-in-advance constraint $\hat{p}_t C_t = 1/\varphi_t$ ties consumption to real balances, so when faster money growth devalues the cash carried into the period, consumption must fall one-for-one with the inflation tax, $\tilde{C}_t = -\tilde{\varphi}_t - \tilde{p}_t$. The low intertemporal elasticity of substitution ($1/\eta = 0.667$) does not prevent this jump; it governs how the household copes with it over time, making it reluctant to let consumption swing and so returning it smoothly to the stationary state rather than overshooting. It is also the CES curvature that makes the money condition (13) forward-looking, so that expected future inflation, not just current inflation, moves today's price and consumption.

Second, indivisible labor makes the real side respond strongly. Because the felicity is linear in hours, the marginal disutility of work is constant, so labor supply is highly elastic: when the inflation tax makes cash-financed consumption expensive, the household readily substitutes toward work, and hours rise on impact. This is the channel through which a purely monetary disturbance moves output, the rental rate and the wage at all. In a divisible-labor economy, where the disutility of work is convex, this reallocation would be damped.

Third, the seigniorage structure decides *what* the tax falls on. Money is needed only for consumption, not for investment, so the inflation tax hits the cash good and spares capital accumulation; the household therefore shifts away from consumption and toward hours and, initially, investment. As



(a) Nominal block: money growth, price, consumption.



(b) Real allocation: capital, hours, rental, wage.

Figure 1: Impulse responses to a positive money-growth (seigniorage) shock, in percent deviations from the stationary state.

the deficit shock fades ($\pi = 0.48$) the money growth rate returns to normal and the inflation tax is lifted, but capital, having moved, carries the economy back only slowly, which is the persistent tail visible in Figure 1b.

5 Summary

I solved a Hansen indivisible-labor economy with cash-in-advance and seigniorage in which consumption enters utility through a CES term with inelastic intertemporal elasticity of substitution ($\eta = 1.5$, so $1/\eta = 0.667$). I set out the household, firm and government problems, derived the first-order conditions — isolating the single money condition that carries η — and computed the stationary state. I then log-linearized the model with Uhlig’s method, arranged the coefficients into the canonical matrices A through N , and solved for the recursive law of motion P, Q, R, S with `llinsolve.m`. Finally, using `impres.m`, I computed and graphed the impulse responses to a positive money-supply growth-rate shock. The shock acts as an inflation tax: it sharply lowers cash-financed consumption, raises hours, and gradually reduces capital, with all variables returning to the stationary state as the shock fades. I also documented that the model is determinate only for $\eta < \eta^* \approx 1.76$, which is why an inelastic value below that threshold was used.

5.1 MATLAB Code

The numerical solution relies on three routines from the course material. `llinsolve.m` implements Uhlig’s log-linear method, computing the reduced-form matrices P, Q, R, S from the structural matrices A through N by a generalized eigenvalue decomposition. `impres.m` takes those matrices together with the shock persistence matrix N and generates the impulse responses to a unit innovation in a chosen shock. `mpsolve.m`, the logarithmic ($\eta = 1$) cash-in-advance program with seigniorage, was used as the template for the structure of the code and as the benchmark against which the CES steady state was checked. The script below sets the calibration, computes the stationary state, builds the matrices, calls `llinsolve.m`, and plots the impulse responses.

```

1 % =====
2 % Hansen economy with CES utility and seigniorage
3 % Final Exam - Dynamic Stochastic Models (Prof. McCandless)
4 % CES curvature eta = 1.5 (inelastic: elasticity 1/eta = 0.667)
5 % Solves the log-linear model (Uhlig's method) and gives P,Q,R,S
6 % =====
7 clear
8 clc
9 close all
10
11 % ----- 1. Calibration (book Sec. 8.9, with eta = 1.5) -----
12 theta = 0.36; % capital share
13 beta = 0.99; % discount factor (quarterly)

```

```

14 delta = 0.025;      % depreciation rate
15 gamma = 0.95;      % persistence of the technology shock
16 pie   = 0.48;      % persistence of the deficit (money) shock
17 varphi = 1.3180;    % steady-state gross money growth rate (phi bar)
18 eta   = 1.5;      % CES curvature; elasticity 1/eta = 0.667 (inelastic)
19 BB    = -2.5805;    % indivisible-labor coefficient (BB, not B: B is a matrix)
20
21 % ----- 2. Stationary state -----
22 rbar   = 1/beta - 1 + delta;      % capital FOC
23 KoverH = (theta/rbar)^(1/(1-theta)); % capital-labor ratio
24 wbar   = (1-theta)*(theta/rbar)^(theta/(1-theta)); % wage
25 pbar   = (-BB/(wbar*beta))^(1/eta)*varphi^(1/eta-1); % normalized price (money
      FOC)
26 Cbar   = 1/(pbar*varphi);      % consumption (CIA)
27 gbar   = (1 - 1/varphi)/pbar;  % government spending
28 Hbar   = (1/pbar)/(wbar + (rbar-delta)*KoverH); % hours (resource constraint)
29 Kbar   = KoverH*Hbar;          % capital
30 Ybar   = Kbar^theta * Hbar^(1-theta); % output
31 mbar   = 1;                    % normalized money
32
33 fprintf('\n==== Stationary state (eta = %.2f) =====\n', eta);
34 fprintf('rbar   = %10.6f\n', rbar);
35 fprintf('KoverH = %10.6f\n', KoverH);
36 fprintf('wbar   = %10.6f\n', wbar);
37 fprintf('pbar   = %10.6f\n', pbar);
38 fprintf('Cbar   = %10.6f\n', Cbar);
39 fprintf('gbar   = %10.6f\n', gbar);
40 fprintf('Hbar   = %10.6f\n', Hbar);
41 fprintf('Kbar   = %10.6f\n', Kbar);
42 fprintf('Ybar   = %10.6f\n', Ybar);
43
44 % ----- 3. Canonical-form matrices -----
45 % States x = [K(t+1); phi(t)] ; Jumps y = [r; w; p; H] ; Shocks z = [lambda; g]
46 A = [ 0          -1/varphi
47       Kbar        0
48       0           0
49       0           0 ];
50
51 B = [ 0          0
52       -(rbar+1-delta)*Kbar  0
53       1-theta          0
54       -theta           0 ];
55
56 C = [ 0          0          pbar*gbar  0
57       -rbar*Kbar -wbar*Hbar -1/pbar   -wbar*Hbar
58       1          0          0        -(1-theta)
59       0          1          0         theta ];

```

```

60
61 D = [ 0      pbar*gbar
62       0      0
63      -1      0
64      -1      0 ];
65
66 F = [ 0      0
67       0      eta-1 ];
68
69 G = zeros(2,2);
70 H = zeros(2,2);
71
72 J = [ beta*rbar  -1      0      0
73       0           0      eta-1  0 ];
74
75 K = [ 0  1  0  0
76       0  1  1  0 ];
77
78 L = zeros(2,2);
79 M = zeros(2,2);
80
81 N = [ gamma  0
82       0      pie ];
83
84 % ----- 4. Solve (Uhlig's method, course routine) -----
85 % unitr = 0 : keep only strictly stable roots (|eigenvalue| < 1)
86 [P,Q,R,S] = llinsolve(A,B,C,D,F,G,H,J,K,L,M,N,0);
87
88 fprintf('\n==== Solution matrices =====\n');
89 fprintf('P ='); disp(P)
90 fprintf('Q ='); disp(Q)
91 fprintf('R ='); disp(R)
92 fprintf('S ='); disp(S)
93
94 % ----- 5. Impulse responses to the money-growth shock -----
95 runlgth = 60;           % length of the impulse response
96 shock    = 2;           % 2 = deficit / money-growth shock (2nd column of N)
97
98 % impres returns z = [x ; y] with rows [K(t+1); phi; r; w; p; H]
99 z = impres(P,Q,R,S,N,1:6,shock,runlgth);
100 close(gcf);             % close the raw figure that impres draws automatically
101
102 Ktil    = z(1,:);        % capital          K(t+1)
103 phitil  = z(2,:);        % money growth  phi
104 rtil    = z(3,:);        % rental rate  r
105 wtil    = z(4,:);        % wage         w
106 ptil    = z(5,:);        % norm. price  p

```

```

107 Htil    = z(6,:);           % hours           H
108 Ctil    = -phitil - ptil;    % consumption from log-linear CIA:  p + C = -phi
109
110 t = 1:runlgth;
111
112 palette = [0.00 0.32 0.60;    % blue
113            0.85 0.33 0.10;    % orange
114            0.47 0.67 0.19;    % green
115            0.49 0.18 0.56];   % purple
116
117 % ----- Figure 1: nominal block (money growth, price, consumption) -----
118 figure(1)
119 set(gcf,'Color','white','ToolBar','none','MenuBar','none')
120 hold on
121 plot(t, 100*phitil, 'Color',palette(1,:), 'LineWidth',1.8)
122 plot(t, 100*ptil,   'Color',palette(2,:), 'LineWidth',1.8)
123 plot(t, 100*Ctil,   'Color',palette(3,:), 'LineWidth',1.8)
124 plot([t(1) t(end)],[0 0],'Color',[0.6 0.6 0.6],'LineWidth',0.8)
125 hold off
126 ax = gca;
127 set(ax,'Color','white','FontName','Helvetica','FontSize',11,'Box','off')
128 grid on; grid minor
129 ytickformat('%0.0f%%')
130 xlabel('Periods')
131 ylabel('Deviation from steady state')
132 legend('\phi (money growth)','p (price)','C (consumption)', ...
133        'Location','best','Box','off')
134 title('Money-growth shock: nominal block','FontWeight','bold','FontSize',13)
135 exportgraphics(gcf,'irf_nominal.pdf','ContentType','vector')
136
137 % ----- Figure 2: real allocation (capital, hours, rental, wage) -----
138 figure(2)
139 set(gcf,'Color','white','ToolBar','none','MenuBar','none')
140 hold on
141 plot(t, 100*Ktil, 'Color',palette(1,:), 'LineWidth',1.8)
142 plot(t, 100*Htil, 'Color',palette(2,:), 'LineWidth',1.8)
143 plot(t, 100*rtil, 'Color',palette(3,:), 'LineWidth',1.8)
144 plot(t, 100*wtil, 'Color',palette(4,:), 'LineWidth',1.8)
145 plot([t(1) t(end)],[0 0],'Color',[0.6 0.6 0.6],'LineWidth',0.8)
146 hold off
147 ax = gca;
148 set(ax,'Color','white','FontName','Helvetica','FontSize',11,'Box','off')
149 grid on; grid minor
150 ytickformat('%0.1f%%')
151 xlabel('Periods')
152 ylabel('Deviation from steady state')
153 legend('K_{t+1} (capital)','H (hours)','r (rental)','w (wage)', ...

```

```

154         'Location','best','Box','off')
155 title('Money-growth shock: real allocation','FontWeight','bold','FontSize',13)
156 exportgraphics(gcf,'irf_real.pdf','ContentType','vector')

```

The two course routines called by the script are reproduced below, exactly as provided in the course material.

```

1 function [P,Q,R,S]=llinsolve(A,B,C,D,F,G,H,J,K,L,M,N,unitr)
2 %set unitr=1 if there is a unit root in the model. Any other value assumes
3 %that all |eigenvalues|<1. Error message is sent if number of acceptable
4 %eigenvalues does not equal the number of state variables (and
5 %expectational equations).
6 [numj nums]=size(A);
7 I1=eye(nums);
8 Z1=zeros(nums);
9 invC=inv(C);
10 psy=F-J*invC*A;
11 lambda=J*invC*B-G+K*invC*A;
12 T=K*invC*B-H;
13 AA1=[lambda T
14      I1 Z1];
15 AA2=[psy Z1
16      Z1 I1];
17 [eigvec eigval]=eig(AA1,AA2);
18 diageigval=diag(eigval);
19 if unitr==1
20     iz=find(abs(diageigval)<=1.00001);
21 else
22     iz=find(abs(diageigval)<1.);
23 end
24 if length(iz)~=nums
25     diageigval
26     error('incorrect number of stable eigenvalues')
27     return
28 end
29 DD=diag(diageigval(iz));
30 ei=size(eigvec);
31 EE=eigvec(ei/2+1:ei,iz);
32 P=EE*DD*inv(EE);
33 R=-invC*(A*P+B);
34 I2=eye(size(N));
35 QQ=kron(N',(F-J*invC*A))+kron(I2,(J*R+F*P+G-K*invC*A));
36 invQQ=inv(QQ);
37 QQQ=((J*invC*D-L)*N+K*invC*D-M);
38 [aa,bb]=size(QQQ);
39 Qfindvert=QQQ(:);
40 Qvert=invQQ*Qfindvert;

```

```

41 Q=reshape(Qvert,aa,bb);
42 S=-invC*(A*Q+D);

```

```

1 function z=impres(P,Q,R,S,N,vecvar,shock,runlgth)
2 %generates impulse response functions from the output of llinsolve and N
3 [aq,numshocks]=size(Q);
4 [numjump,numstate]=size(R);
5 e=zeros(numshocks,runlgth);
6 e(shock,2)=1;
7 xold=zeros(numstate,1);
8 shockold=zeros(numshocks,1);
9 for i=1:runlgth
10     shocknew=N*shockold+e(:,i);
11     x(:,i)=P*xold+Q*shocknew;
12     y(:,i)=R*xold+S*shocknew;
13     xold=x(:,i);
14     shockold=shocknew;
15 end
16 z= [x
17     y];
18 z=z(vecvar,:);
19 plot(z')

```